

Advanced Flying Robots – INDI Controller

Current Progress: Motion Planning + INDI on Crazyflie 2.1

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Motion Planning – INDI Controller

Motion planning pipeline

- Minimum-snap polynomial trajectories; two timing modes: Mode 1 (k_t timing), Mode 3 (time redistribution).
- Differential flatness maps the planned trajectory to the controller feedforward (thrust, attitude, $\dot{\omega}_d$).

INDI controller

- Incremental, sensor-based control law (Tal & Karaman 2020).
- Benchmarked on the CF2.1: 37% better tracking than geometric. Port to the thrust-upgraded / brushless drone is ongoing (Next Steps).

Planning Modes 1 And 3

Both modes share the same minimum-snap polynomial backbone and flatness-based attitude; they differ only in how the segment times T_i are chosen.

Mode	Timing idea	Segment-time rule
1	k_t -based allocation	$T_i = \max(d_i/v_{avg}, T_{\min}), \quad v_{avg} = 0.5 + 1.5\sqrt{k_t}$
3	Time redistribution	$T_i^{\text{tgt}} = T_{\text{tot}} \frac{J_i}{\sum_j J_j}$ (clamped to $T_{\min}$); then $T_i \leftarrow T_i + \frac{1}{2}(T_i^{\text{tgt}} - T_i)$, starting from Mode 1

T_i : time of segment i ; $d_i = \|p_{i+1} - p_i\|$: segment length; v_{avg} : average speed; k_t : speed / aggressiveness scalar; $T_{\min} = 0.15$ s: minimum segment time; J_i : snap cost of segment i ; T_{tot} : total trajectory time.

Minimum Snap — Segment Polynomial & Cost

What we minimize

$$\sigma_i(\hat{t}) = \sum_{k=0}^8 c_{i,k} \hat{t}^k, \quad \hat{t} = \frac{\tau}{T_i} \in [0, 1], \quad J = \sum_i \frac{1}{T_i^7} \int_0^1 \left(\frac{d^4 \sigma_i}{d\hat{t}^4} \right)^2 d\hat{t}$$

σ_i	position polynomial, segment i (per axis)
i	segment index
k	polynomial power (0–8)
$c_{i,k}$	coefficients (the unknowns solved for)
$\hat{t}$	normalized time, $0 \rightarrow 1$ per segment
τ	real time; $\hat{t} = \tau / T_i$
T_i	duration of segment i [s]
$d^4 \sigma_i / d\hat{t}^4$	snap (4th derivative of position)
$1/T_i^7$	normalized→real time scaling
J	total snap cost — the objective (sum of snap ² over all segments)

Minimum Snap — Quadratic Program

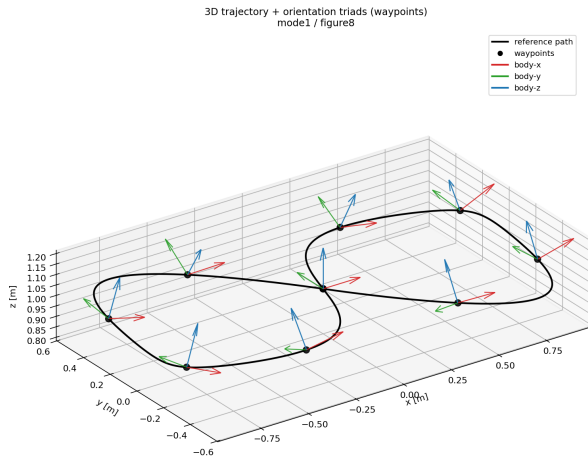
QP form

$$\min_c \frac{1}{2} c^\top Q c + q^\top c \quad \text{s.t.} \quad A_{eq} c = b_{eq}$$

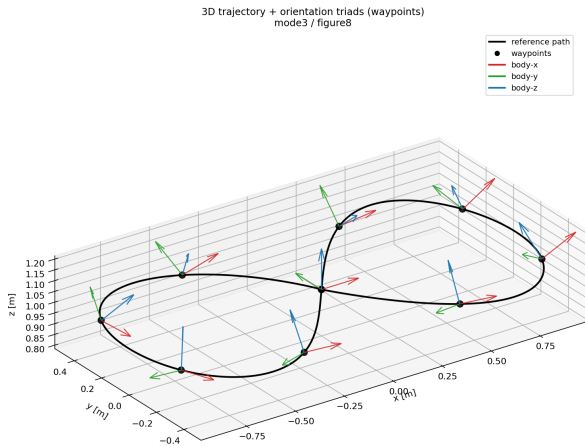
Equality constraints: waypoint interpolation $\sigma_i(0) = w_i$, $\sigma_i(1) = w_{i+1}$; derivative continuity $\frac{1}{T_i^r} \sigma_i^{(r)}(1) = \frac{1}{T_{i+1}^r} \sigma_{i+1}^{(r)}(0)$, $r = 1, \dots, 4$. Solved per axis; coupling re-enters through flatness; thrust / rate / torque limits checked *after* solving.

c	stacked vector of all coefficients $c_{i,k}$	
Q	cost (Hessian) matrix encoding snap	$c^\top Q c$: quadratic cost ($= J$)
$q, q^\top c$	linear cost (zero for pure min-snap)	$\frac{1}{2}$: conventional QP factor
A_{eq}, b_{eq}	equality-constraint matrix / right-hand side	
w_i	waypoint position (segment start / end)	
$\sigma_i^{(r)}, r$	r -th derivative of segment poly; order 1 vel, 2 acc, 3 jerk, 4 snap	
$1/T_i^r$	r -th-derivative time scaling	

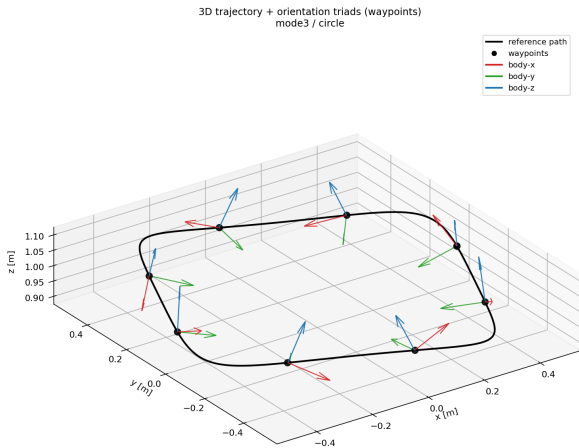
Mode 1 — Planned Figure-8 (3D Trajectory)



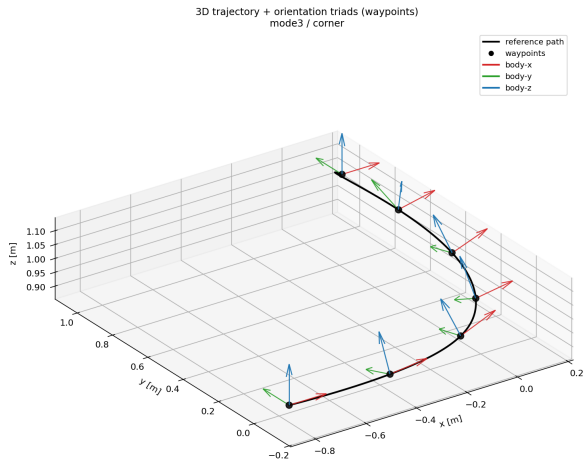
Mode 3 — Planned Figure-8 (3D Trajectory)



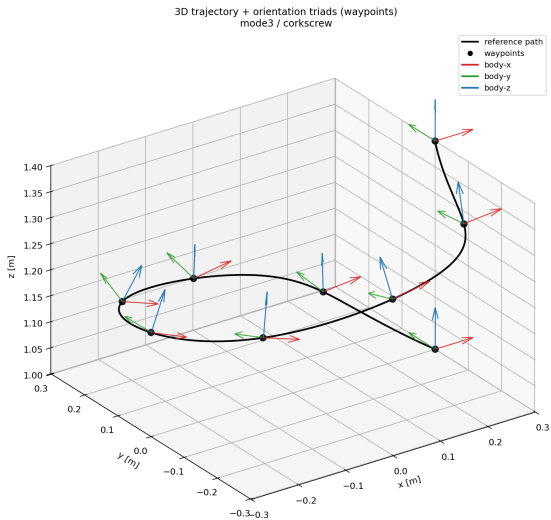
Mode 3 — Circle (3D Trajectory)



Mode 3 — Corner (3D Trajectory)

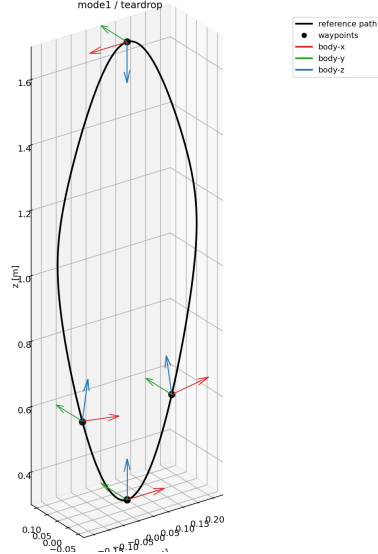


Mode 3 — Corkscrew (3D Trajectory)



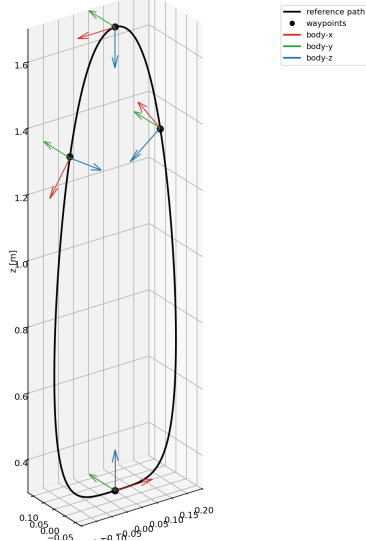
Teardrop — Round Bottom (3D Trajectory)

3D trajectory + orientation triads (waypoints)
model1 / teardrop



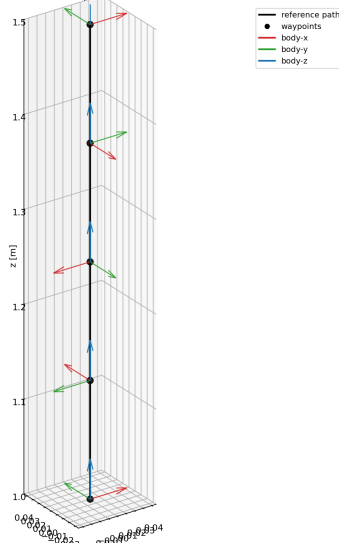
Teardrop — Round Top (3D Trajectory)

3D trajectory + orientation triads (waypoints)
model / teardrop_high

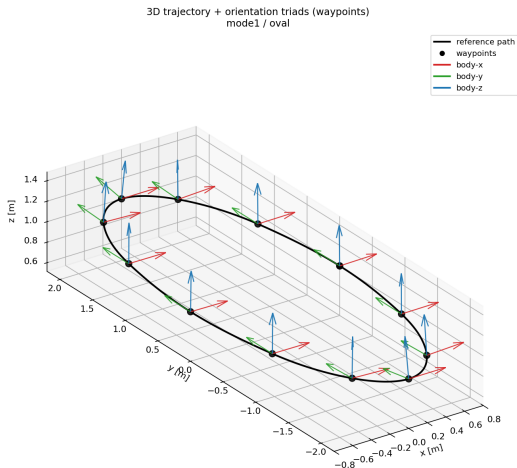


Mode 3 — Screw (3D Trajectory)

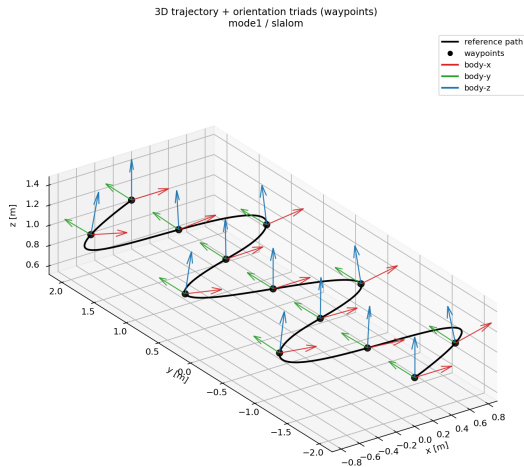
3D trajectory + orientation triads (waypoints)
mode3 / screw



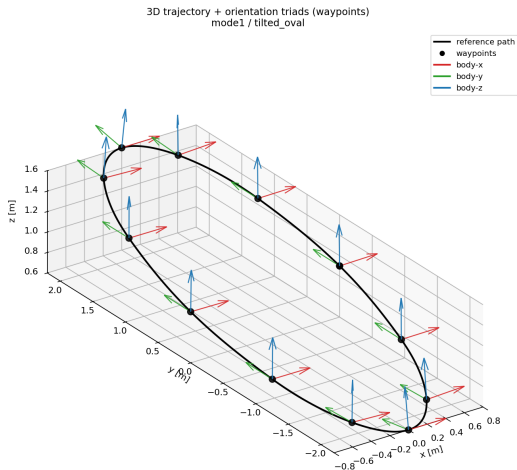
Racing Trajectory — Oval (3D Trajectory)



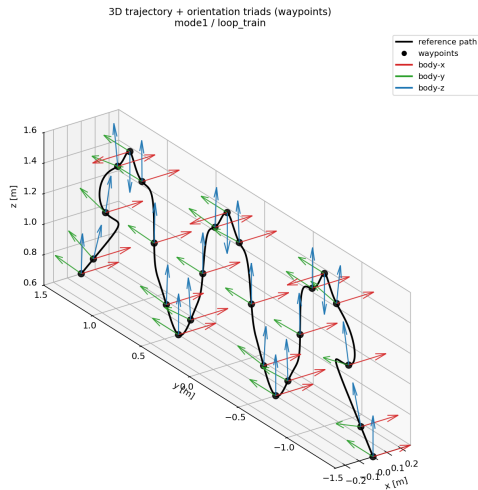
Racing Trajectory — Slalom (3D Trajectory)



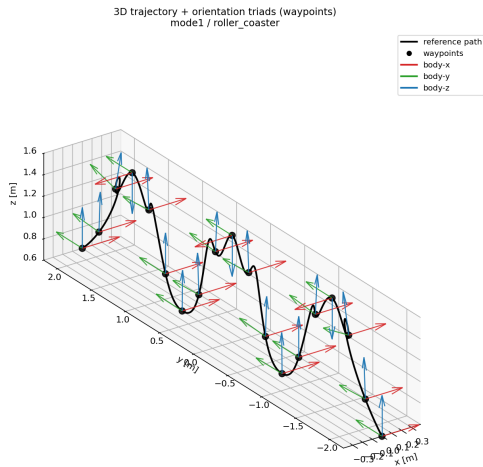
Racing Trajectory — Tilted Oval (3D Trajectory)



Brushless Aerobatic — Loop Train (3D Trajectory)



Brushless Aerobatic — Roller-Coaster (3D Trajectory)



Why INDI? Incremental, Sensor-Based Control

Geometric (model-based) attitude law

$$\tau = -K_R e_R - K_\omega e_\omega + \omega \times J\omega$$

where e_R : attitude error, e_ω : body-rate error, K_R, K_ω : torque gains, J : inertia.

Needs an accurate model (inertia, aerodynamics, actuator mapping). Model error $\rightarrow$ tracking error.

INDI idea — Incremental Nonlinear Dynamic Inversion

- Instead of the full model torque, measure the current angular acceleration + actuator torque and command only the *increment* needed to reach the desired angular acceleration.
- The measured $\dot{\omega}$ already contains the true dynamics + disturbances $\Rightarrow$ model error largely cancels.
- Applied to *both* loops: torque increment (attitude) and acceleration increment (position).

Reference: Tal & Karaman, “Accurate Tracking of Aggressive Quadrotor Trajectories”, 2020.

INDI Control Law — Outer Loop (Position)

$$e_p = p_d - p, \quad e_v = v_d - v$$

position / velocity error

$$a_{\text{meas}} = R a_{\text{imu}} - g e_3$$

measured accel (world)

$$a_{\text{model}} = \frac{1}{m} \left(\sum_i k_{t,i} \text{rpm}_i^2 \right) z_B - g e_3$$

modeled thrust accel

$$f_d = a_d + K_P e_p + K_V e_v + g e_3 + (a_{\text{meas}} - a_{\text{model}})$$

← INDI term

$$T = m f_d \cdot z_B$$

collective thrust

p, v	measured position, velocity	R	attitude (body→world rotation)
subscript d	trajectory reference (p_d, v_d, a_d)	z_B	body-up axis (3rd column of R)
e_p, e_v	position, velocity error	a_{imu}	accelerometer (specific force)
a_d	feedforward acceleration	$k_{t,i}$	motor- i thrust constant
K_P, K_V	position / velocity gains (xy, z)	rpm_i	motor- i speed
m	mass g : gravity	$a_{\text{meas}}, a_{\text{model}}$	measured vs. modeled accel
e_3	world up-axis (0, 0, 1)	f_d, T	force command; collective thrust

INDI Control Law — Inner Loop (Attitude)

$$\begin{aligned}e_R &= \frac{1}{2}(R_d^\top R - R^\top R_d)^\vee, & e_\omega &= \omega - \omega_d & \text{attitude / rate error} \\ \dot{\omega}_{\text{ref}} &= \dot{\omega}_d - k_r e_R - k_w e_\omega & & & \text{desired angular accel} \\ \tau &= \tau_{\text{cur}} + J(\dot{\omega}_{\text{ref}} - \dot{\omega}_{\text{meas}}) & & & \leftarrow \text{INDI term}\end{aligned}$$

R, R_d current, desired attitude
 $(\cdot)^\vee$ 'vee': skew matrix $\rightarrow$ 3-vector
 e_R attitude (orientation) error
 ω, ω_d measured (gyro), desired body rate
 e_ω body-rate error
 $\dot{\omega}_d$ feedforward angular accel

k_r, k_w INDI inner gains [$\text{s}^{-2}, \text{s}^{-1}$]
 $\dot{\omega}_{\text{ref}}$ desired angular acceleration
 $\dot{\omega}_{\text{meas}}$ measured ang. accel (next slide)
 τ_{cur} current torque from RPM (next slide)
 J inertia τ : torque command

$\dot{\omega}_{\text{ref}}, \dot{\omega}_{\text{meas}}$ share one Butterworth (matched phase); $k_r, k_w \neq$ geometric torque gains. Geometric mode = INDI terms off
 $\Rightarrow \tau \rightarrow -K_R e_R - K_\omega e_\omega + \omega \times J\omega$.

INDI: Angular Acceleration And Control Effectiveness

Measured angular acceleration $\dot{\omega}_{\text{meas}}$ (the critical signal)

$$\dot{\omega}_{\text{meas}} = \text{LP} \left(\frac{\omega_k - \omega_{k-1}}{\Delta t} \right), \quad \text{2nd-order Butterworth, cutoff } f_{bw}$$

ω_k : current gyro sample, Δt : control period, f_{bw} : filter cutoff. Trade-off: lower $f_{bw} \rightarrow$ less noise but more phase lag; higher $\rightarrow$ faster but noisier.

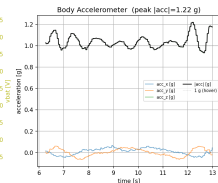
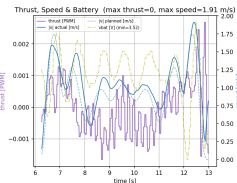
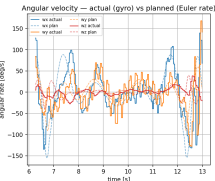
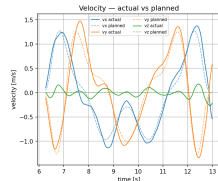
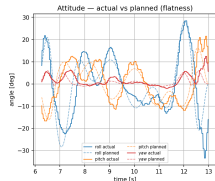
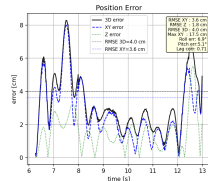
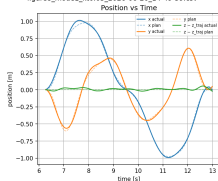
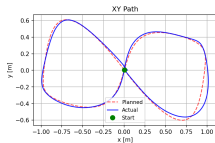
Current torque τ_{cur} from motor speeds

$$f_i = k_{t,i} \text{rpm}_i^2, \quad \tau_{\text{cur}} = \begin{bmatrix} \ell (f_3 + f_4 - f_1 - f_2) \\ \ell (f_2 + f_3 - f_1 - f_4) \\ c_{Q/T} (f_2 + f_4 - f_1 - f_3) \end{bmatrix}$$

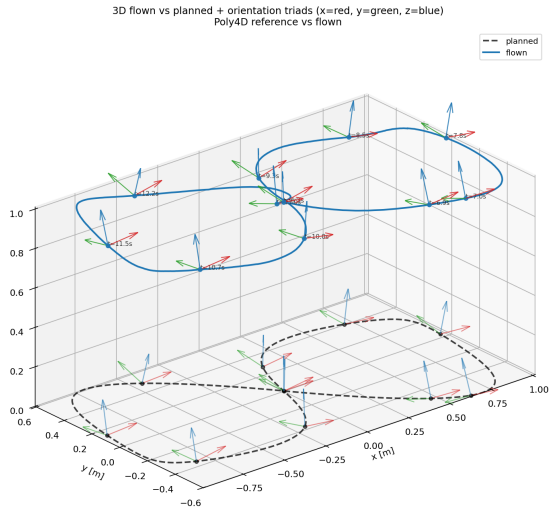
f_i : thrust of motor i , $k_{t,i}$: per-motor thrust constant, rpm_i : motor speed (RPM deck), ℓ : projected arm length, $c_{Q/T}$: torque-to-thrust ratio. τ_{cur} and $\dot{\omega}_{\text{meas}}$ feed the inner-loop increment.

INDI Flight: Best Figure-8 — Full Analysis

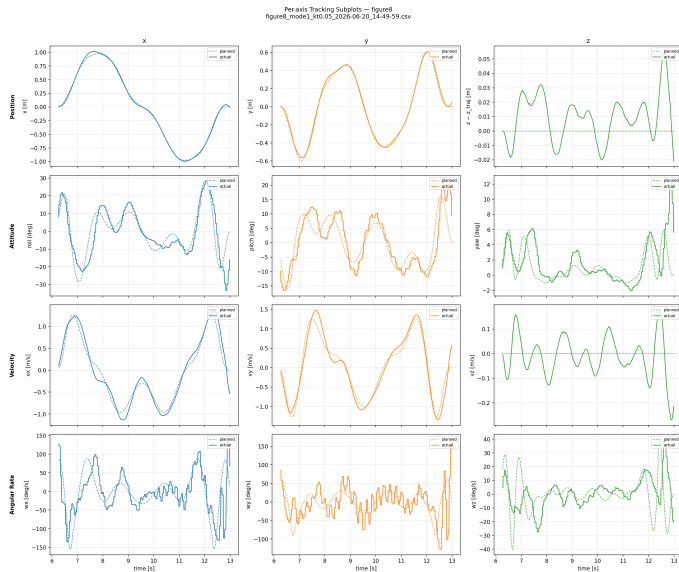
Trajectory Analysis — figure8 (SPEED_SCALE=1.0, XY_SCALE=1.0)
figure8_model1_kt05_2026-06-20_14-49-59.csv



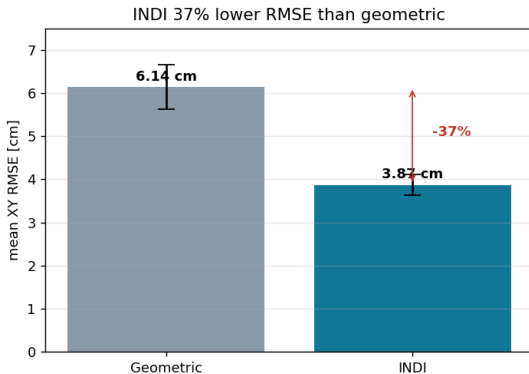
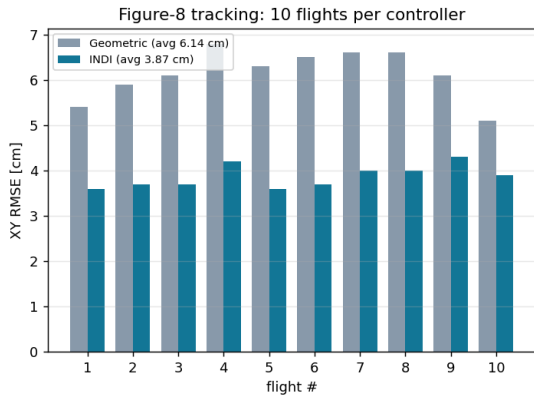
INDI Flight: Best Figure-8 — 3D Orientation



INDI Flight: Best Figure-8 — Per-Axis Tracking



Geometric vs. INDI — 10 Flights Each



INDI Speed Sweep — Figure-8 Aggressiveness Limit

k_t	Lap time [s]	XY RMSE	Status
0.01	8.84	2.8 cm	stable
0.02	8.07	3.4 cm	stable
0.03	7.57	4.7 cm	stable
0.04	7.19	4.0 cm	stable
0.05	6.88	5.4 cm	stable — best tracking
0.10	5.90	11.3 cm	stable — degrading
0.15	5.32	16.0 cm	stable — speed limit
0.20	4.91	—	crash — genuine instability

- ① INDI on the thrust-upgraded drone
- ② INDI on the brushless platform
- ③ Maximize figure-8 speed while stable
- ④ Fly more different trajectories